

DETERMINANT-POSITIVE H^1 OUTERS AND THE BLECHER–LABUSCHAGNE EXTREMAL QUESTION

FA_BANACH.001 LITERATURE-IMPLIED PACKET

STATUS

This is a literature-implied partial answer to a question in Blecher–Labuschagne, *Applications of the Fuglede–Kadison determinant: Szego’s theorem and outers for noncommutative H^p* , arXiv:math/0609662. The source paper asks on page 14:

Is there a characterization of outers in H^1 in terms of extremals, as in the deLeeuw–Rudin theorem?

Of course in the case that $\mathcal{D}$ is finite dimensional one can drop the word ‘strongly’ in the last several results. In particular, in the case that the algebra A is antisymmetric, these results and their proofs are much simpler and are spelled out in our survey [7].

Question. Is there a characterization of outers in H^1 in terms of extremals, as in the deLeeuw–Rudin theorem of e.g. [17] p. 139–142], or [13] pp. 137–139]?

Acknowledgements. We thank Barry Simon for helpful information on Szegő’s theorem and its generalizations, and for pointing out Verblunsky’s precedence to the result usually attributed to Kolmogorov and Krein. We thank W. B. Arveson for continual encouragement. Finally, we are grateful to Q. Xu for suggesting several years ago to look at the noncommutative variant of the Riesz–Szegő theorem; and

The later paper by Bekjan and Xu, *Riesz and Szego type factorizations for noncommutative Hardy spaces*, arXiv:0705.1947, gives such a characterization in the determinant-positive subcase.

SUPPORTING THEOREM

Let A be a finite subdiagonal algebra in a finite von Neumann algebra $\mathcal{M}$, let $H^p(A)$ be the associated noncommutative Hardy space, let $\Phi : \mathcal{M} \rightarrow \mathcal{D} = A \cap A^*$ be the conditional expectation, and let Δ denote the Fuglede–Kadison determinant.

Bekjan–Xu Corollary 4.11 states that if $0 < p \leq \infty$ and $h \in H^p(A)$ satisfies $\Delta(h) > 0$, then h is outer if and only if for every $x \in H^p(A)$ with $|x| = |h|$ one has

$$\Delta(\Phi(x)) \leq \Delta(\Phi(h)).$$

Corollary 4.11 *Let $0 < p \leq \infty$ and $h \in H^p(\mathbf{A})$ with $\Delta(h) > 0$. Then h is outer iff for any $x \in H^p(\mathbf{A})$ with $|x| = |h|$ we have $\Delta(\Phi(x)) \leq \Delta(\Phi(h))$.*

Proof. Assume h outer. Then by Corollary 2.3 and Theorem 4.4

$$\Delta(\Phi(x)) \leq \Delta(x) = \Delta(h) = \Delta(\Phi(h)).$$

Conversely, let $h = uk$ be the decomposition given by Theorem 4.8 with k outer. Then

$$\Delta(h) = \Delta(k) = \Delta(\Phi(k)) \leq \Delta(\Phi(h));$$

so h is outer by Theorem 4.4 □

Corollary 4.12 *Let $0 < p, q, r \leq \infty$ such that $1/p = 1/q + 1/r$. Let $x \in H^p(\mathbf{A})$ be such that $\Delta(x) > 0$. Then there exist $y \in H^q(\mathbf{A})$ and $z \in H^r(\mathbf{A})$ such that*

$$x = yz \quad \text{and} \quad \|x\|_p = \|y\|_q \|z\|_r.$$

Proof. This proof is similar to that of Theorem 3.4. Instead of Theorem 3.1 we now use Theorem 4.8. Indeed, by the later theorem, we can find a unitary $u_2 \in \mathbf{M}$ and an outer $h_2 \in H^{p/r}(\mathbf{A})$ such that $|x|^{p/r} = u_2 h_2$. Once more applying this theorem to $v|x|^{p/q} u_2$, we have a similar factorization: $v|x|^{p/q} u_2 = u_1 h_1$, where v is the unitary in the polar decomposition of x . Since h_1 and h_2 are outer, we deduce, as in the proof of Corollary 4.11, that $x = v h$ and $z = h$, giving

Taking $p = 1$ gives a determinant-extremal characterization of strongly outer H^1 functions: among analytic competitors with the same modulus, the outer element is exactly the one whose diagonal expectation has maximal Fuglede–Kadison determinant.

IDENTIFICATION AND SCOPE

This answers the Blecher–Labuschagne question for the determinant-positive part of H^1 . The relation is an agent-identified implication rather than an explicitly named resolution in Bekjan–Xu: the supporting paper cites and extends the Blecher–Labuschagne outer theory, and Corollary 4.11 specializes directly to the $p = 1$ extremal characterization requested by the source.

The packet is not a full answer to all nearby open points. It does not remove the hypothesis $\Delta(h) > 0$, does not handle determinant-zero outers, does not give a literal deLeeuw–Rudin extreme-point theorem, and does not settle the separate question whether every outer element of H^1 is the square of an outer element of H^2 .

SEARCH NOTES

The local parsed arXiv corpus was searched for the exact square-of-outer phrase, for the deLeeuw–Rudin/extremal wording, and for determinant/outer variants. The exact square question reappears in a Blecher–Labuschagne survey arXiv:math/0611879, but no later full closure was found in the local corpus. ArXiv web searches for the same determinant and noncommutative Hardy keywords identified Bekjan–Xu arXiv:0705.1947 as the relevant later source.

REFERENCES

- [1] D. P. Blecher and L. E. Labuschagne, *Applications of the Fuglede-Kadison determinant: Szego’s theorem and outers for noncommutative H^p* , arXiv:math/0609662.
- [2] T. N. Bekjan and Q. Xu, *Riesz and Szego type factorizations for noncommutative Hardy spaces*, arXiv:0705.1947.